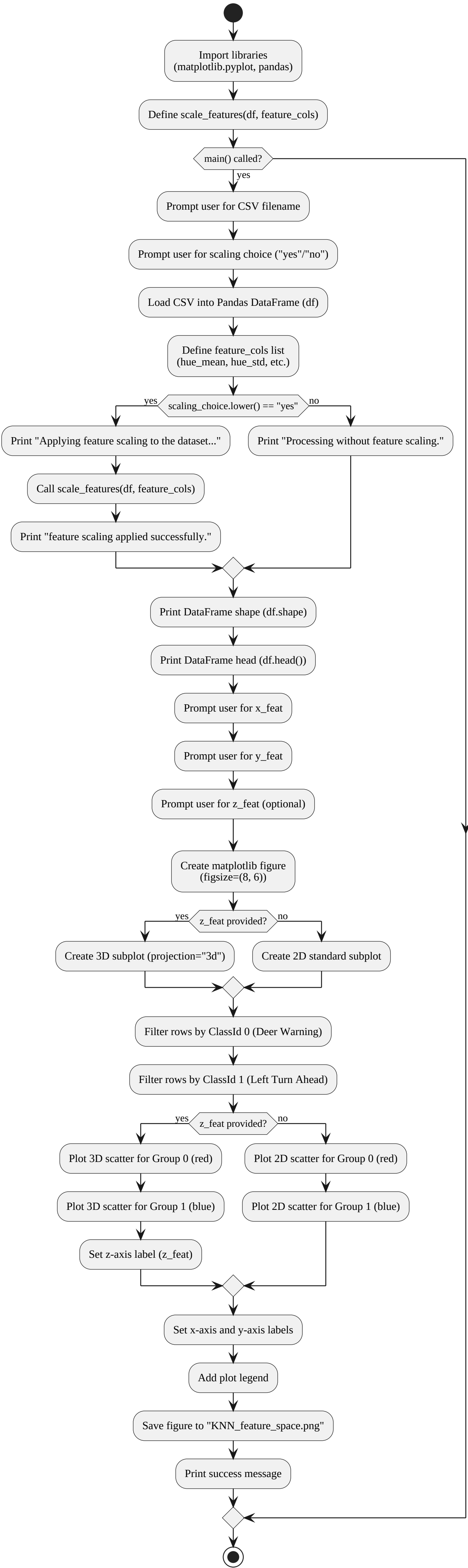
Import libraries (matplotlib.pyplot, pandas)
Define scale_features(df, feature_cols)
main() called?
yes
Prompt user for CSV filename
Prompt user for scaling choice ("yes"/"no")
Load CSV into Pandas DataFrame (df)
Define feature_cols list (hue_mean, hue_std, etc.)
yes
scaling_choice.lower() == "yes"
no
Print "Applying feature scaling to the dataset..."
Call scale_features(df, feature_cols)
Print "feature scaling applied successfully."
Print "Processing without feature scaling."
Print DataFrame shape (df.shape)
Print DataFrame head (df.head())
Prompt user for x_feat
Prompt user for y_feat
Prompt user for z_feat (optional)
Create matplotlib figure (figsize=(8, 6))
yes
z_feat provided?
no
Create 3D subplot (projection="3d")
Create 2D standard subplot
Filter rows by ClassId 0 (Deer Warning)
Filter rows by ClassId 1 (Left Turn Ahead)
yes
z_feat provided?
no
Plot 3D scatter for Group 0 (red)
Plot 3D scatter for Group 1 (blue)
Set z-axis label (z_feat)
Plot 2D scatter for Group 0 (red)
Plot 2D scatter for Group 1 (blue)
Set x-axis and y-axis labels
Add plot legend
Save figure to "KNN_feature_space.png"
Print success message